



Product Manual | ICX-FieldHawk

Handheld Real-Time Spectrum Analyzer

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OVERVIEW

The ICX series is a USB-based real-time spectrum analyzer built around a high-performance superheterodyne front end with up to 14 stages of preselection filtering. This architecture is intended to deliver strong noise performance, wide dynamic range, and good immunity to nearby interferers, while keeping analysis speed high and system cost manageable.

The compact, rugged enclosure is designed to simplify integration into space- and cost-constrained test setups without sacrificing RF performance such as spectral purity, making the platform a practical fit for portable and embedded monitoring applications.

Every model in the ICX-FieldHawk family shares a common API, so a test program written for one model carries over to another in the line without code changes. The API supports C/C++, C#, Python, MATLAB, Qt, and LabVIEW, on both Windows and Linux.

Standard measurement functions include channel power, occupied bandwidth (OBW), X dB bandwidth, harmonic measurement, AM/FM demodulation, and automated phase-noise measurement.

Key Features

- High-performance superheterodyne receiver with up to 14 stages of preselection filtering
- Frequency range: 9 kHz to 4.5 / 6 / 9 GHz, depending on model
- Displayed average noise level (DANL): -168 dBm/Hz at 1 GHz; -167 dBm/Hz at 9 GHz
- Phase noise: < -110 dBc/Hz at 10 kHz offset (1 GHz carrier)
- Phase-noise measurement mode: < -125 dBc/Hz at 10 kHz offset
- Analysis bandwidth: 50 MHz standard, 100 MHz with Option 50
- Sweep speed: > 1 THz/s (RBW = 250 kHz)
- IF and image rejection: > 95 dBc typical
- Built-in GNSS supporting frequency calibration
- USB 3.0 / 2.0 Type-C interface
- Rich APIs and example programs across major languages and platforms
- Windows 11/10/8/7 (x86, x64); Debian 12/11/10 (x64, AArch64); Ubuntu 24.04/22.04/20.04/18.04 (x64, AArch64)
- Standard SCPI command support

SPECICX-GEN3 CONTROL SOFTWARE

SpecICX-gen3 is the companion control application for the ICX series. It groups measurement setup, mode selection, and live results into one workspace, and offers nine primary operating modes: Standard Spectrum Analysis, IQ Streaming, Power Detection Analysis, Real-Time Spectrum Analysis, Phase Noise Measurement, Basic Vector Modulation Analysis (Option 71), Pulse Analysis (Option 72), Harmonic Analysis, and Mapping.

Standard Spectrum Analysis Mode

Provides a full-span sweep together with common measurements such as channel power, OBW, ACPR, IM3, and spectral emission mask (SEM). Recording and playback of captured spectra are supported, and auxiliary tools — signal tracking, a peak table, and amplitude correction — combine into a single workspace for routine spectrum checks.

IQ Streaming Mode

Captures IQ data at up to 100 MHz of analysis bandwidth using several trigger methods. The mode displays IQ time-domain waveforms alongside spectrum and spectrogram views, and includes AM/FM demodulation and digital down-conversion (DDC).

Power Detection Analysis Mode

Analyzes time-domain signal behavior within the active analysis bandwidth, which suits applications centered on in-band power-versus-time relationships, such as characterizing pulsed signals.

Real-Time Spectrum Analysis Mode

Runs on a high-speed, FPGA-based FFT engine with gapless, non-overlapping frame processing, giving continuous real-time visibility across the full analysis bandwidth.

Basic Vector Modulation Analysis Mode (Option 71)

Demodulates and analyzes 2ASK, 2FSK, 4FSK, GMSK, BPSK, QPSK, 8PSK, and 16/64/128/256QAM signals, reporting constellation and modulation-quality measurements.

Harmonic Analysis Mode

Detects and measures up to 10 harmonic components, reporting harmonic peak levels, per-harmonic channel power, and total harmonic distortion.

Phase Noise Measurement Mode

Covers offset ranges from 1 Hz to 10 MHz for carrier phase-stability evaluation, with an automatic carrier-search routine that locates the target carrier without manual tuning.

Pulse Analysis Mode (Option 72)

Measures pulsed signals with pulse widths of 32 ns or greater, reporting top/base level, droop, overshoot, ripple, rise/fall time and edge, width, PRI, and duty cycle.

Mapping Mode

Combines GPS position data with measurement results to build heat maps and related visualizations — channel power, occupied bandwidth, time, coordinates, altitude, pixel position, and azimuth — useful for radio-monitoring and interference-hunting work in the field.

MEASUREMENT FUNCTIONS

Beyond the primary operating modes, SpecICX-gen3 includes a standard toolset of measurement and workflow aids common across every mode:

- Channel Power — total power within a defined channel bandwidth
- OBW — occupied bandwidth for a percentage of total power
- ACPR — adjacent channel power ratio
- IM3 — third-order intermodulation measurement
- SEM — spectral emission mask pass/fail evaluation
- AM/FM Demodulation — modulation depth, deviation, and audio-domain analysis
- Antenna Factor — correction curve applied for calibrated field-strength measurements
- Amplitude Offset — external attenuation/gain compensation
- Signal Track — automatic peak tracking across sweeps
- Peak Table — sortable list of detected signal peaks
- Data Record and Playback — capture and replay of spectrum data
- Multiple Unit Display — simultaneous multi-unit view for distributed monitoring

SPECIFICATIONS

Frequency

Parameter	ICX-045	ICX-060	ICX-090
Frequency range	9 kHz to 4.5 GHz	9 kHz to 6 GHz	9 kHz to 9 GHz
Reference clock	Internal or external; manual correction or GNSS calibration available		
Frequency accuracy — TCXO (std.)	< 0.5 ppm, manual correction available		
Frequency accuracy — OCXO (Option 01)	< 0.2 ppm, manual correction available		
Frequency correction via GNSS	< 0.05 ppm when GNSS is locked		
Aging & temperature stability — TCXO (std.)	< 1 ppm/year, < 1 ppm		
Aging & temperature stability — OCXO (Option 01)	< 1 ppm/year, < 0.15 ppm		
Built-in GNSS 1PPS accuracy	± 100 ns		

Spectrum Purity

SSB Phase Noise (dBc/Hz), Guaranteed / Typical

Carrier frequency	1 GHz	4.5 GHz	6 GHz	9 GHz
100 Hz	-85 / -88	-78 / -81	-75 / -78	-74 / -77
1 kHz	-97 / -99	-90 / -93	-86 / -89	-88 / -91
10 kHz	-110 / -112	-103 / -106	-100 / -103	-99 / -102
100 kHz	-114 / -115	-104 / -107	-104 / -107	-99 / -102
1 MHz	-131 / -132	-119 / -122	-118 / -121	-116 / -119

Residual Response (dBm), Guaranteed / Typical

Frequency range (RBW = 1 kHz)	R.L. = 0 dBm	R.L. = -50 dBm
9 kHz to 100 MHz	-90 / -94	-115 / -117
100 MHz to 4.5 GHz	-90 / -95	-120 / -128
4.5 GHz to 6 GHz	-90 / -96	-120 / -130
6 GHz to 9 GHz	-90 / -98	-120 / -129

IIP3 / IIP2 (dBm), Typical

Carrier frequency	1 GHz	4.5 GHz	6 GHz	9 GHz
R.L. = 20 dBm	42 / 80	37 / 80	32 / 80	36 / 80
R.L. = 0 dBm	23 / 80	22 / 80	20 / 80	14 / 80
R.L. = -20 dBm	3 / 60	3 / 45	-1 / 45	-5 / 45

Parameter	Specification
Image rejection	> 90 dBc; > 95 dBc typical
IF rejection	> 90 dBc; > 95 dBc typical
Local-oscillator-related spurious	< -65 dBc, center frequency $\pm (N/M) \times 100$ MHz, N,M = 1,2,3,4,5...

Amplitude

Parameter	Specification
Max. input power (CW)	23 dBm (50 MHz to max. frequency, preamplifier off); 10 dBm (9 kHz to 50 MHz, or preamplifier on)
Max. DC voltage	± 10 VDC
Display range	DANL to 30 dBm

Parameter	Specification
Level accuracy	± 1.0 dB, 20-30°C typical conditions; ± 1.5 dB over full temperature range, all conditions
IF in-band flatness	$< \pm 1.5$ dB (50 MHz analysis bandwidth); $< \pm 2.0$ dB (100 MHz analysis bandwidth)
Reference level	-70 dBm to +23 dBm
RF preamplifiers	Auto or manual
VSWR	$< 2.0:1$ nominal, 90 MHz to max. frequency

Displayed Average Noise Level (DANL), dBm/Hz, RBW = 1 kHz, Guaranteed / Typical

Frequency range	ICX-045		ICX-060		ICX-090	
	-20 dBm	-70 dBm	-20 dBm	-70 dBm	-20 dBm	-70 dBm
9 kHz to 1 MHz	-131 / -134	-141 / -144	-131 / -134	-141 / -144	-131 / -134	-141 / -144
1 MHz to 100 MHz	-151 / -154	-163 / -166	-151 / -154	-163 / -166	-151 / -154	-163 / -166
100 MHz to 3.0 GHz	-141 / -144	-165 / -168	-141 / -144	-165 / -168	-141 / -144	-165 / -168
3.0 GHz to 4.5 GHz	-142 / -145	-165 / -168	-142 / -145	-165 / -168	-142 / -145	-165 / -168
4.5 GHz to 6 GHz	-	-	-143 / -146	-165 / -168	-143 / -146	-165 / -168
6 GHz to 9 GHz	-	-	-	-	-144 / -147	-164 / -167

Standard Spectrum Analysis

Parameter	Specification
Detector	PosPeak, NegPeak, Sample, Average, RMS, RawFrames, MaxPower
RBW	0.1 Hz to 10 MHz
VBW	0.1 Hz to 10 MHz
Data chart	SpecICX-gen3 provides spectrum, spectrogram, and historical-trace views
Measurements	Channel power, OBW, X dB bandwidth, adjacent channel power ratio, IM3
Sweep speed (RBW = 250 kHz, FPGA, B-Nuttall)	≥ 1 THz/s
Sweep speed (RBW = 25 kHz, FPGA, Kaiser)	≥ 300 GHz/s
Sweep speed (RBW = 1 kHz, CPU, B-Nuttall)	≥ 4 GHz/s

IQ Recording

Parameter	Specification
Continuous recording bandwidth	Maximum 50 MHz
Burst recording bandwidth	Maximum 50 MHz / 100 MHz (Option 50); built-in memory depth 128 Mbytes
IQ sample rate	Maximum 62.5 MSPS / 125 MSPS (Option 50); decimate factor 1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, 2048
External trigger response	Maximum 500 times/s

Detection Analysis

Parameter	Specification
Min. time resolution	16 ns / 8 ns (Option 50)
Max. analysis bandwidth	50 MHz / 100 MHz (Option 50)
Detector	PosPeak, NegPeak, Sample, Average, RMS, MaxPower

Real-Time Spectrum Analysis

Parameter	Specification
Max. analysis bandwidth	50 MHz / 100 MHz (Option 50)
Window function	B-Nuttall, Flat-top, LowSideLobe, Rectangle, Kaiser
RBW steps	≥ 12 grades
Amplitude resolution	0.5 dB

Typical FFT Settings

Typical setting	FFT refresh rate, std.	FFT refresh rate, Opt.50	100% POI, std.	100% POI, Opt.50
N = 4096, D = 1	15,258 /s	30,517 /s	131.072 μs	65.536 μs
N = 32, D = 1	1,953,125 /s	3,906,250 /s	1.024 μs	0.512 μs

Phase Noise Measurement

Parameter	Specification
Min. frequency offset	1 Hz

Parameter	Specification
Max. measurement frequency offset	10 MHz
Trace smoothing	Supported

SSB Phase Noise (dBc/Hz), Guaranteed / Typical

Carrier frequency	1 GHz	4.5 GHz	6 GHz	9 GHz
100 Hz	-90 / -94	-80 / -83	-80 / -83	-78 / -81
1 kHz	-113 / -117	-102 / -105	-100 / -103	-95 / -98
10 kHz	-120 / -125	-112 / -115	-109 / -112	-106 / -109
100 kHz	-126 / -129	-114 / -117	-111 / -114	-108 / -111
1 MHz	-136 / -139	-125 / -128	-123 / -126	-119 / -122
10 MHz	-140 / -143	-137 / -140	-137 / -140	-136 / -139

Pulse Detection

Parameter	Specification
Min. pulse width	64 ns / 32 ns (Option 50)
Measurement parameters	Top level, base level, top/base, droop, overshoot, ripple, rise time, rise edge, fall time, fall edge, width, PRI, duty cycle

AM Demodulation

Parameter	Specification
Measurement parameters	Modulation depth, carrier power, modulation rate, signal-to-noise ratio, RMS power, total harmonic distortion, and related parameters
Modulation rate test range	20 Hz to 10 MHz
Modulation rate test accuracy	< 1 Hz when modulation rate < 1 kHz; < 0.1% when modulation rate ≥ 1 kHz
Modulation depth test range & accuracy	5% to 95%, ± 5% nominal

FM Demodulation

Parameter	Specification
Measurement parameters	Modulation frequency offset, carrier power, modulation rate, signal-to-noise ratio, RMS power, total harmonic distortion, and related parameters
Modulation rate test range	20 Hz to 2 MHz
Modulation rate test accuracy	< 1 Hz when modulation rate < 1 kHz; < 0.1% when modulation rate $\geq$ 1 kHz
Frequency offset test range & accuracy	1 kHz to 10 MHz, $\pm$ 6% nominal

Basic Vector Modulation Analysis

Parameter	Specification
Modulation type	ASK: 2ASK — FSK: 2FSK, 4FSK — MSK: GMSK — PSK: BPSK, QPSK, 8PSK — QAM: 16/32/64/128/256QAM
Symbol length	4000 symbols (128QAM, 256QAM); 2000 symbols (others)
Symbol rate	$(1/64 \text{ to } 1/4) \times \text{sample rate}$, $\leq$ 32.5 MSPS
Filter	Root raised cosine
Filter roll-off factor	0.01 to 0.99
Display	Spectrum, constellation, eye diagram, measurement results
Measurement	EVM, amplitude error, phase error, frequency error, signal-to-noise ratio, partial bitstream

General

Parameter	Specification
RF input	N-type female, 50 Ω impedance
Power	USB Type-C, 5 V / 3 A supply capacity; voltage range 4.75-5.25 V; ripple < 200 mVpp
USB port	USB Type-C, USB 3.0 (USB 2.0 bandwidth-limited); requires 5 V / 0.9 A supply
External trigger input	Integrated in Aux I/O, 3.3 V CMOS, high impedance
Trigger output	Integrated in Aux I/O, 3.3 V CMOS
GNSS antenna input	SMA female
Analog IF output	MMCX female, max. output power -25 dBm; 50 Ω impedance, 312.5 MHz $\pm$ 50 MHz

Parameter	Specification
External reference clock input	MMCX female, 10 MHz, amplitude ≥ 1.5 Vpp, 330 Ω impedance
Reference clock output	MMCX female, 100 MHz, amplitude ≥ 0.3 Vpp, programmable on/off
Overall / core weight	≤ 420 g / ≤ 195 g
Overall / core dimensions (L $\times$ W $\times$ H)	$\leq 163 \times 66 \times 37$ mm / $\leq 109 \times 60 \times 16$ mm
Power consumption	≤ 15 W
Packaging accessories	Flash drive $\times 1$, USB 3.0 data cable $\times 1$, USB power cable $\times 1$, power adapter $\times 1$
System requirements	Windows 11/10/8/7 (x86, x64); Debian 12/11/10 (x64, AArch64); Ubuntu 24.04/22.04/20.04/18.04 (x64, AArch64)
Operating/storage temperature — T0 class (std.)	0 to 50°C / -20 to 70°C
Operating/storage temperature — T1 class (Option 40)	-20 to 65°C / -40 to 85°C
Operating/storage temperature — T2 class (Option 41, core only)	-40 to 65°C / -40 to 85°C
Operating relative humidity (ambient $\leq 40^\circ\text{C}$)	5% to 75%
Operating relative humidity (ambient $> 40^\circ\text{C}$)	5% to 45%

Specifications apply under the following conditions: (1) instrument powered on and warmed up for 10 minutes; (2) ambient temperature 25°C; (3) typical and nominal values are not guaranteed and exclude measurement uncertainty; (4) specifications may change with hardware and software revisions without prior notice.

OPTIONS

Code	Description	Type
01	Built-in OCXO reference clock	Built-in hardware
05	Internal high-precision GNSS	Built-in hardware
20	Aux I/O expansion board	Accessory
34	External omnidirectional antenna, 400 MHz-8000 MHz, gain < 2 dBi	Accessory
35	External active directional antenna, 0.5-10 GHz; gain < 5 dBi (amp off), < 25 dBi (amp on)	Accessory
40	T1 temperature class	Built-in hardware
41	T2 temperature class (core only)	Built-in hardware
50	100 MHz analysis bandwidth	Built-in hardware
71	Basic vector modulation analysis	Software
72	Pulse analysis	Software